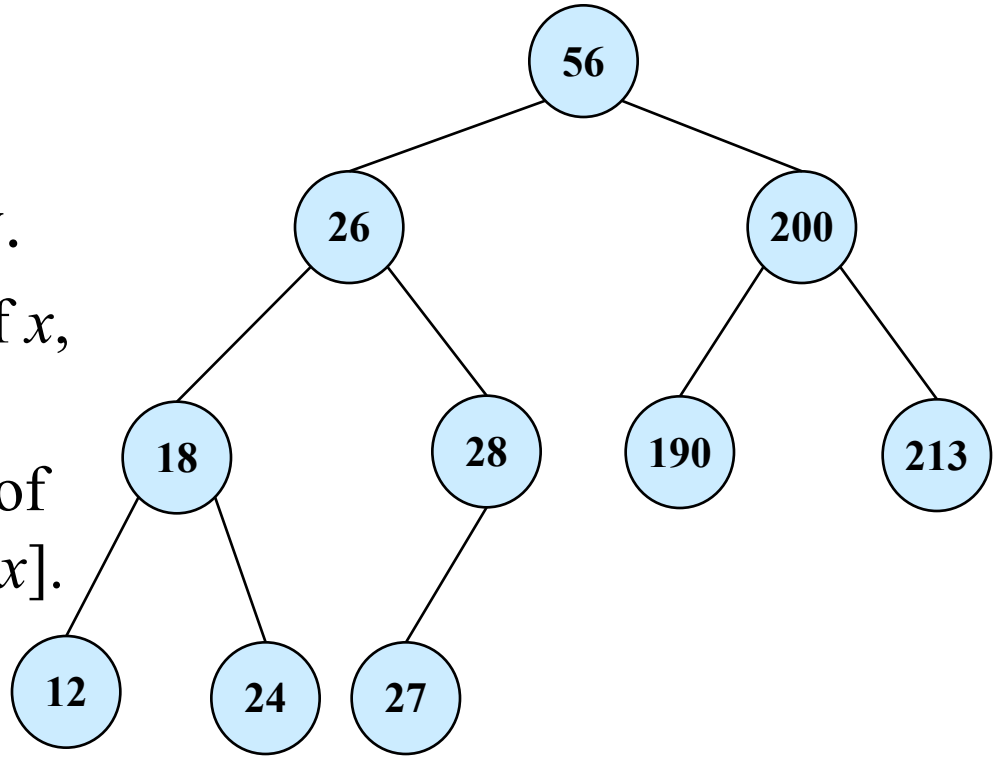
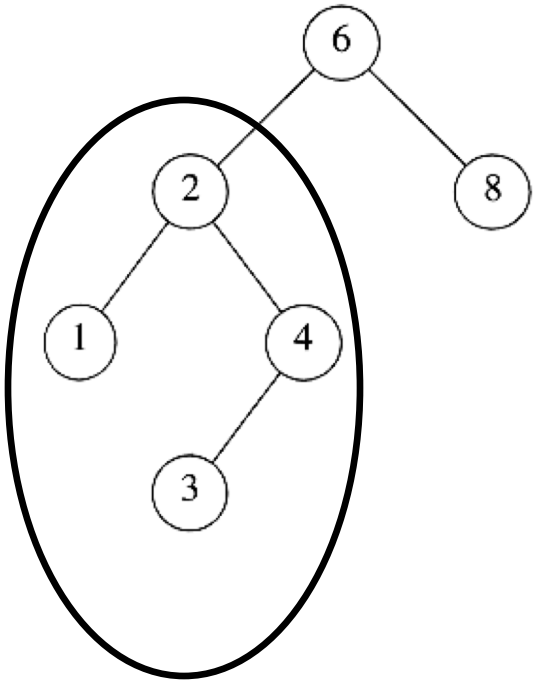
Binary Search Tree

Binary Search Tree Property

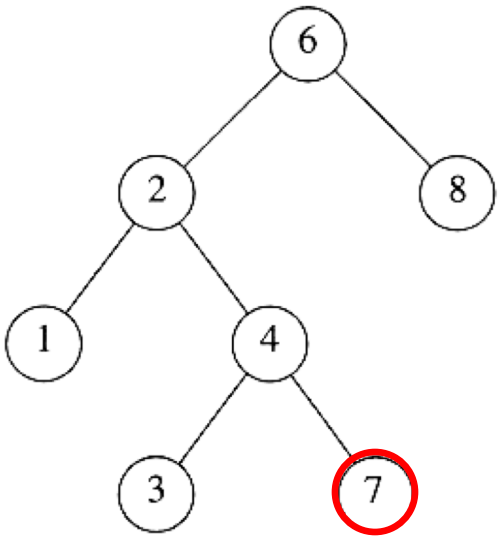
- Stored keys must satisfy the *binary search tree* property.
  - ∀ y in left subtree of x, then $key[y] \leq key[x]$.
  - ∀ y in right subtree of x, then $key[y] \geq key[x]$.



Binary Search Trees



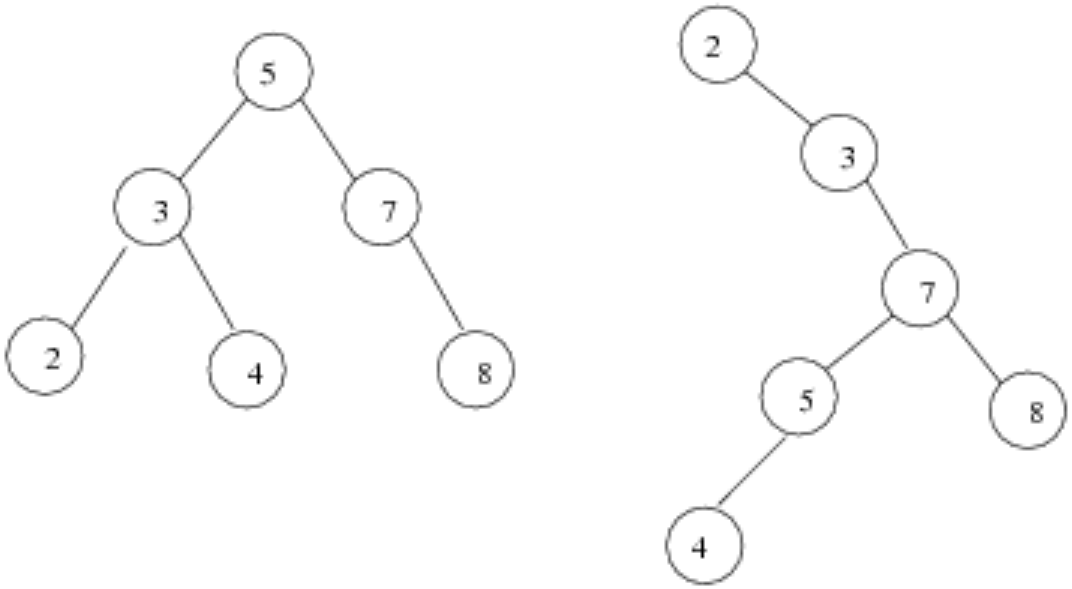
A binary search tree



Not a binary search tree

BST

Two binary search trees representing the same set:

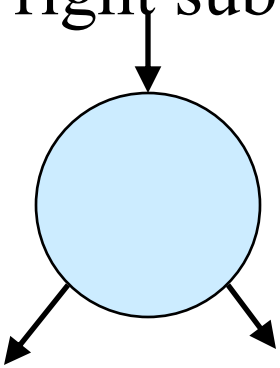


- Average height is O(log N);
- maximum height is O(N)

BST – Representation

- Represented by a linked data structure of nodes.
- *root*(*T*) points to the root of tree *T*.

- Each node contains fields:
  - *key*
  - *left* – pointer to left child: root of left subtree.
  - *right* – pointer to right child : root of right subtree.



Node Structure

```
Struct Tree
{
int data;
Struct Tree *left;
Struct Tree *right;
};
```

Binary Search Trees

- Operations On BST.
  - Search, Minimum, Maximum, Predecessor, Successor, Insert, and Delete.

- Can be used to build
  - Dictionaries.
  - Priority Queues.

Basic Operations

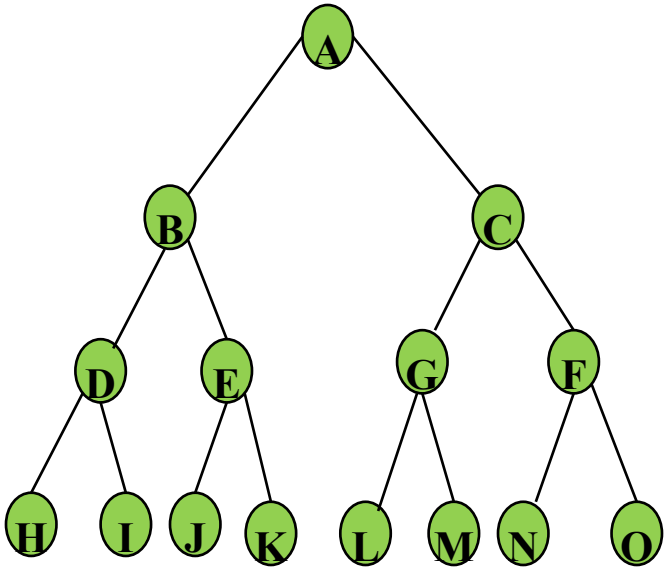
- ✓ Search - compare the values and proceed either to the left or to the right
- ✓ Insertion - unsuccessful search - insert the new node at the bottom where the search has stopped
- ✓ Deletion - replace the value in the node with the smallest value in the right subtree
  or the largest value in the left subtree.
- ✓ Retrieval in sorted order – inorder traversal

Complexity

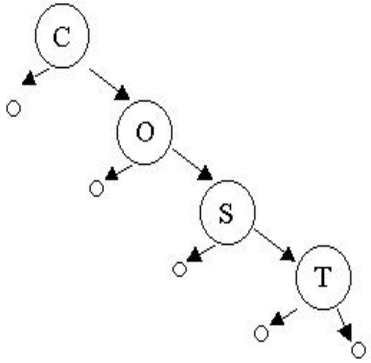
- Logarithmic, depends on the shape of the tree

  In the best case – O(log N) comparisons

  In the worst case – O(N) comparisons



Best case



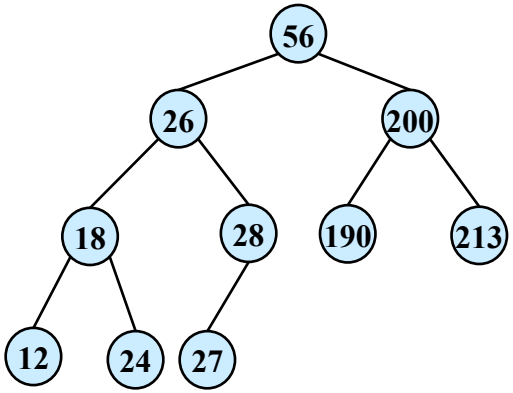
Worst case

Inorder Traversal

The binary-search-tree property allows the keys of a binary search tree to be printed, in (monotonically increasing) order, recursively.

Inorder-Tree-Walk (x)

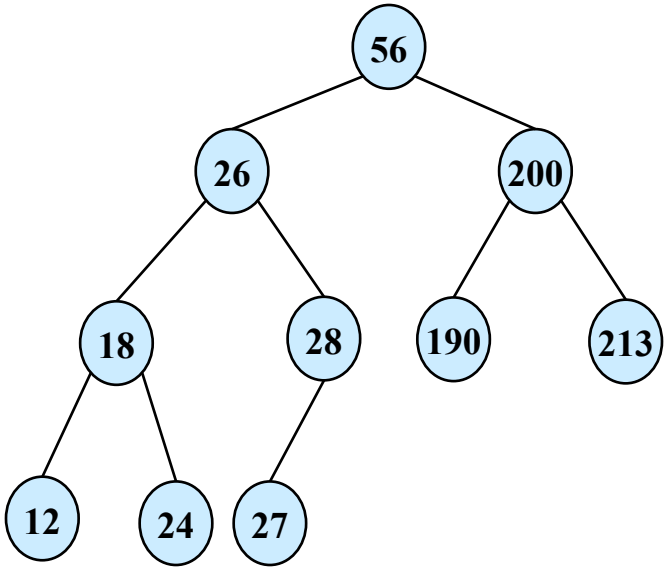
1. **if** $x \neq$ NIL
2. **then** Inorder-Tree-Walk(*left*[*p*])
3. print *key*[*x*]
4. Inorder-Tree-Walk(*right*[*p*])



- **How long does the walk take?**

Tree Search

```
struct Tree *Find(struct Tree *root, int data )
{
If(root ==Null)
    return Null;
If (data<root->data)
    return (Find ( root->left, data));
else if ( data>root->data)
        return (Find ( root->right, data));
return root;
}
```



Finding Min & Max

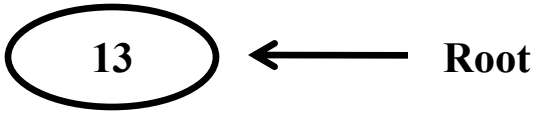
- The binary-search-tree property guarantees that:
  - » The minimum is located at the left-most node.
  - » The maximum is located at the right-most node.

```
Struct Tree *FindMin(Struct Tree *root)
{
if(root ==Null)
    return Null;
else If (root->left==Null)
        return root;
else return FindMin(root->left);
}
```

```
Struct Tree *FindMax(Struct Tree *root)
{
If(root ==Null)
    return Null;
Else If (root->right==Null)
        return root;
Else return FindMax(root->right);
}
```

Insertion in a Binary Search Tree

13, 3, 4, 12, 14, 10, 5, 1, 8, 2, 7, 9, 11, 6, 18



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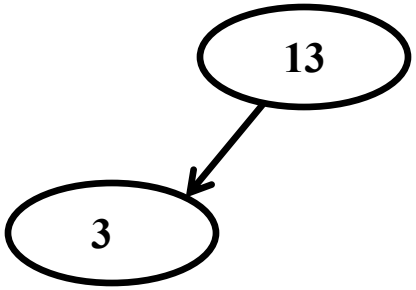

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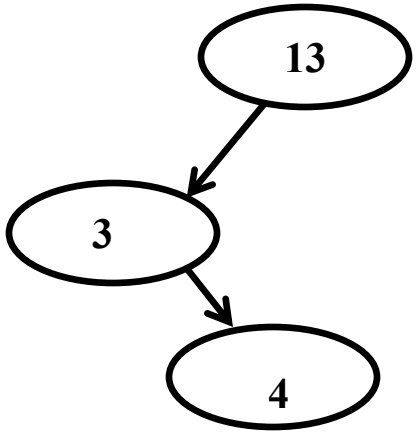

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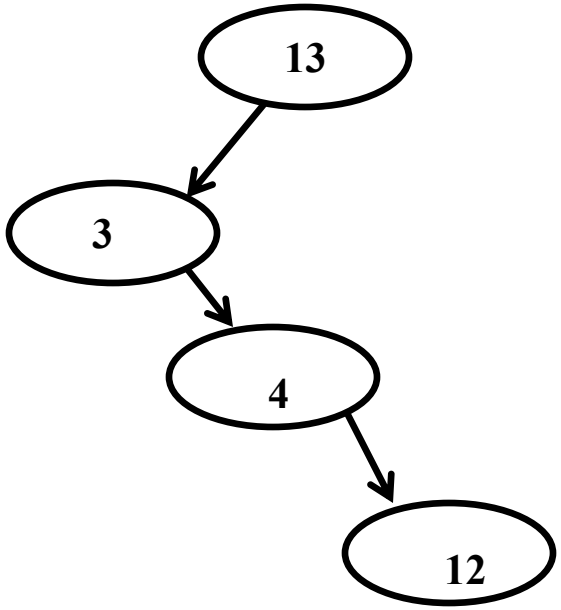

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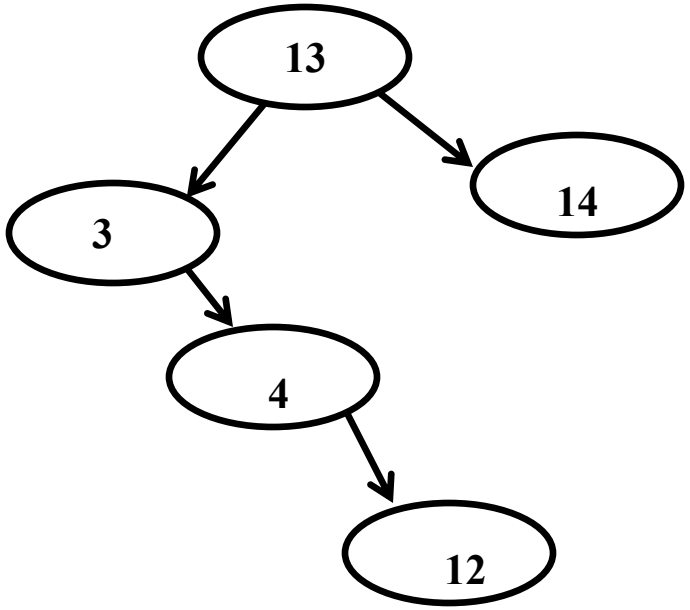

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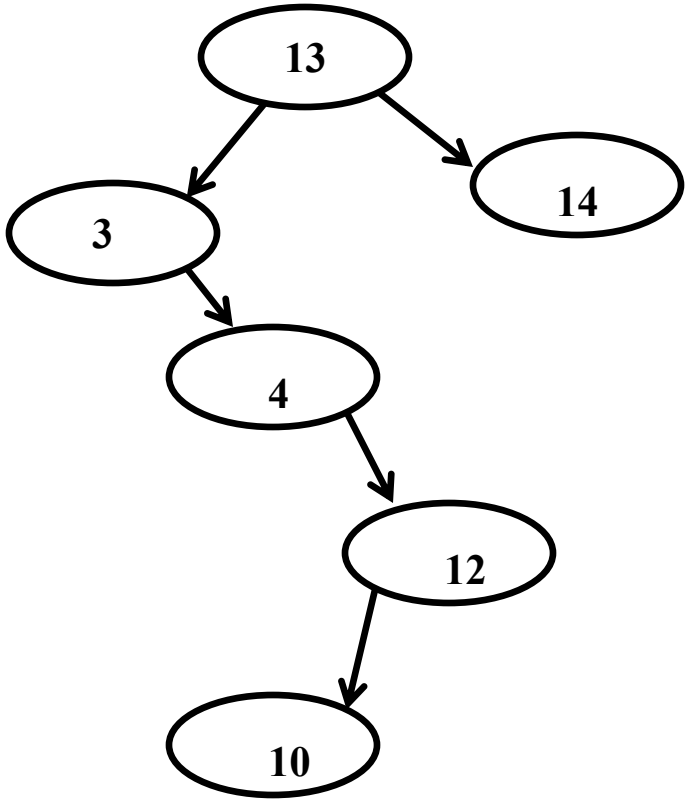

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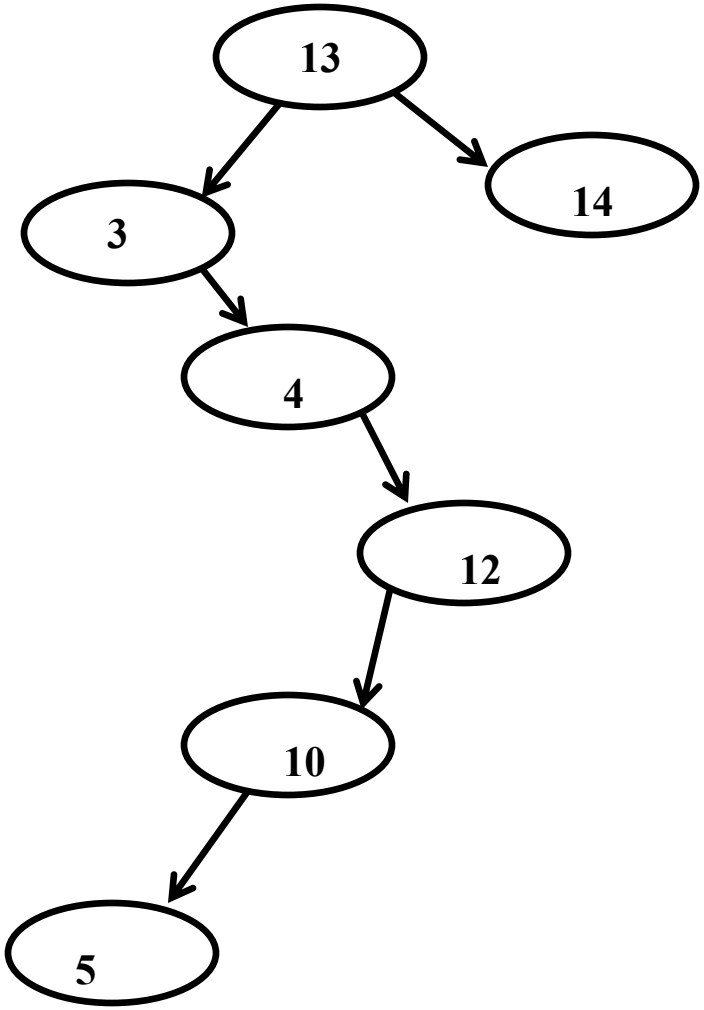

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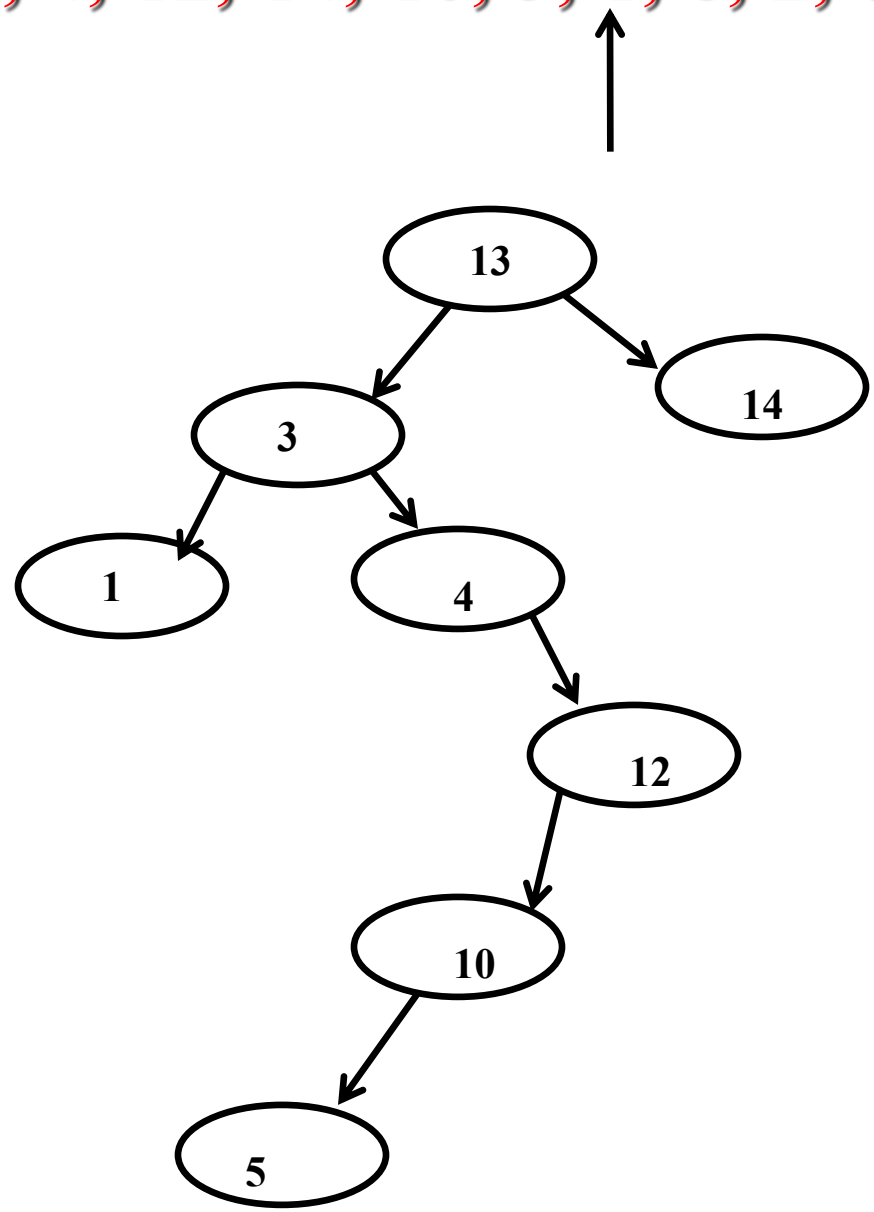

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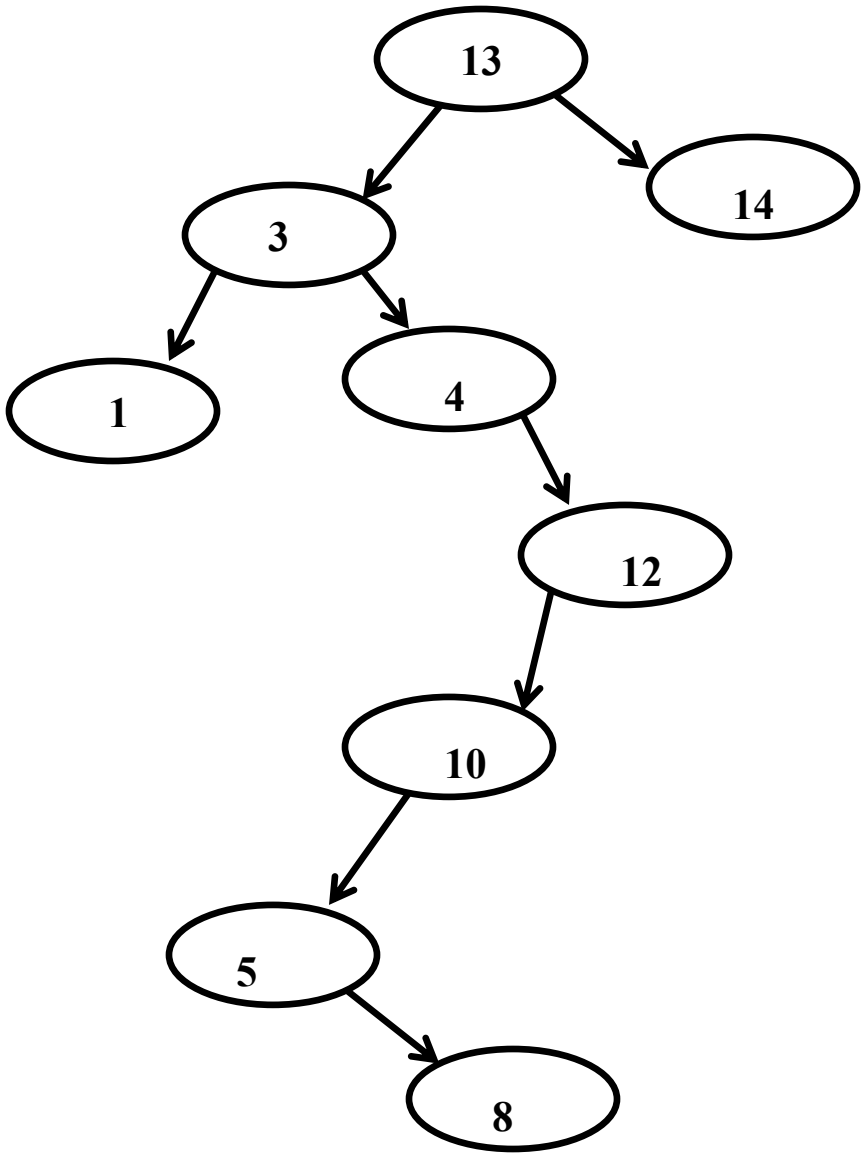

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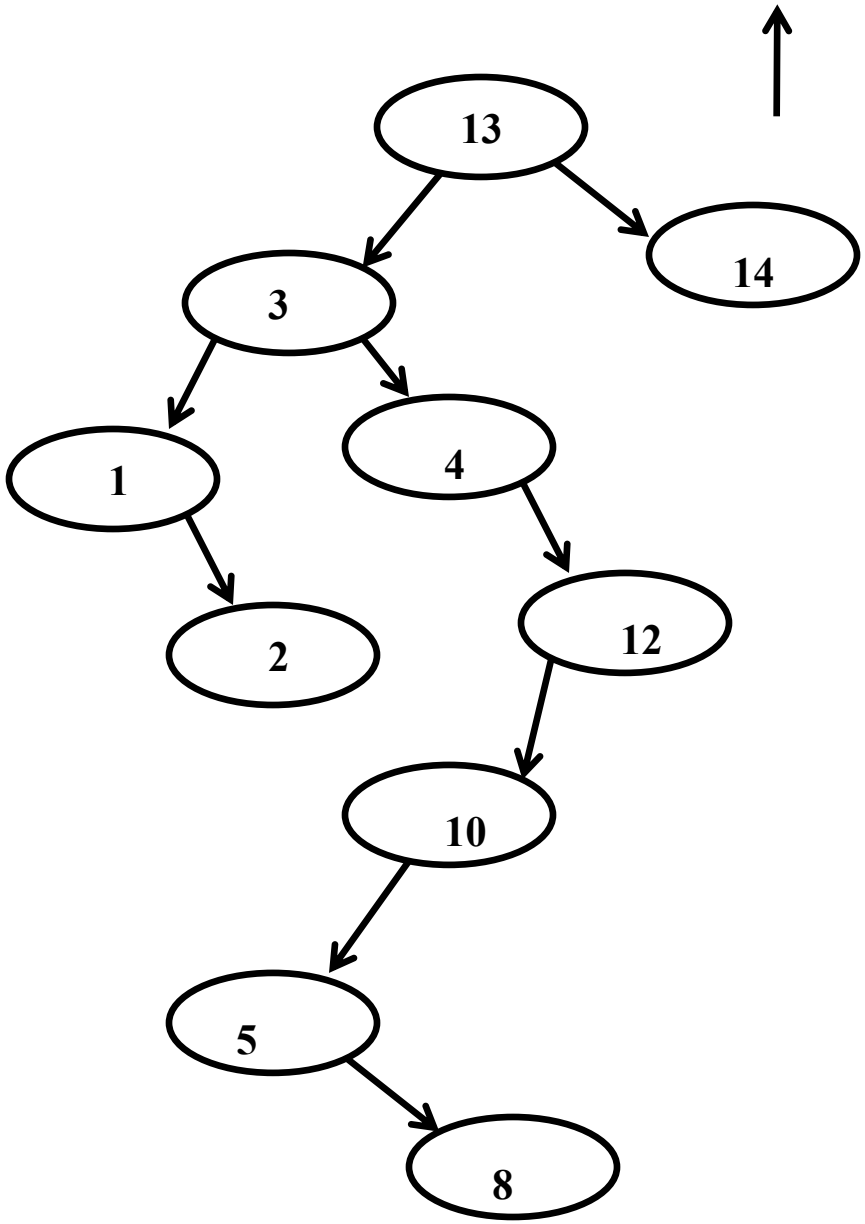

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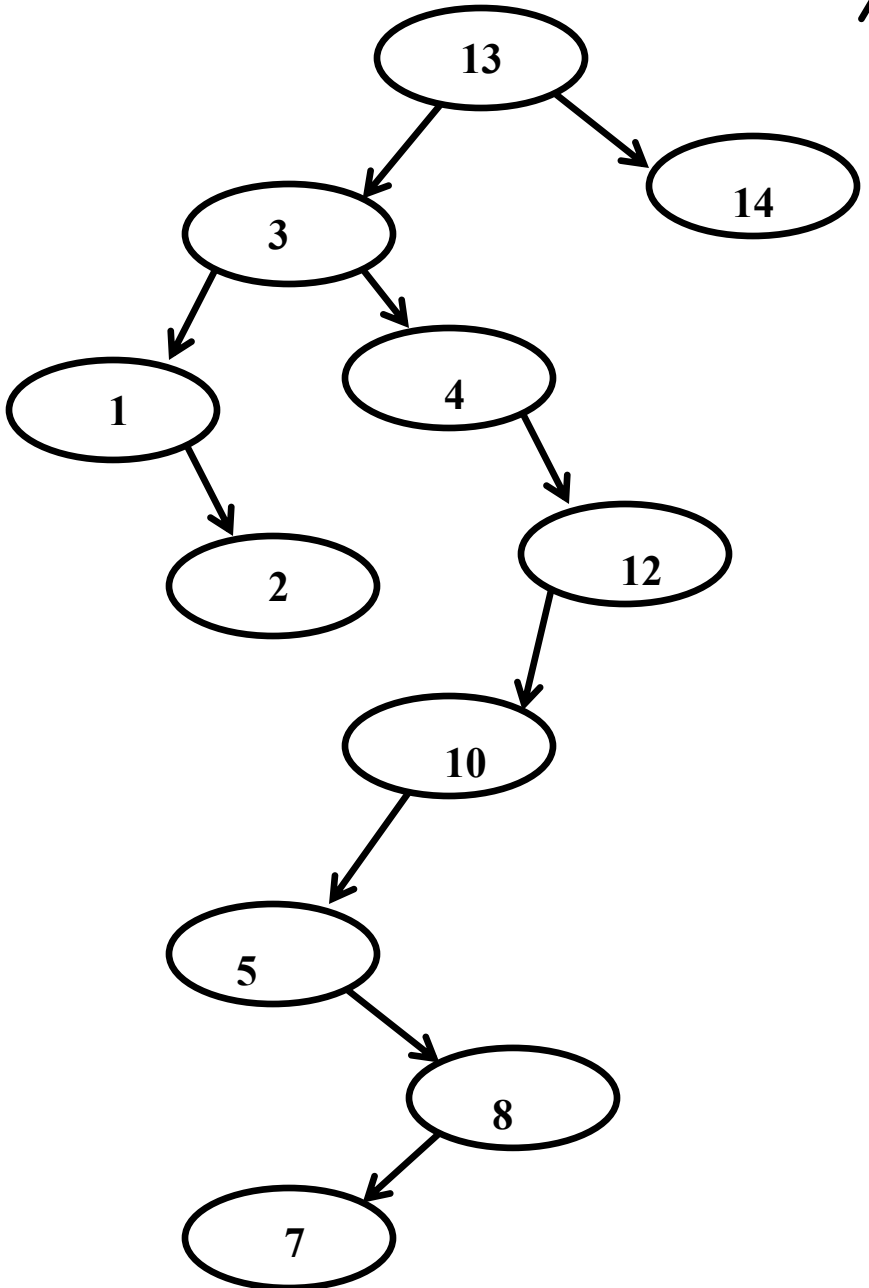

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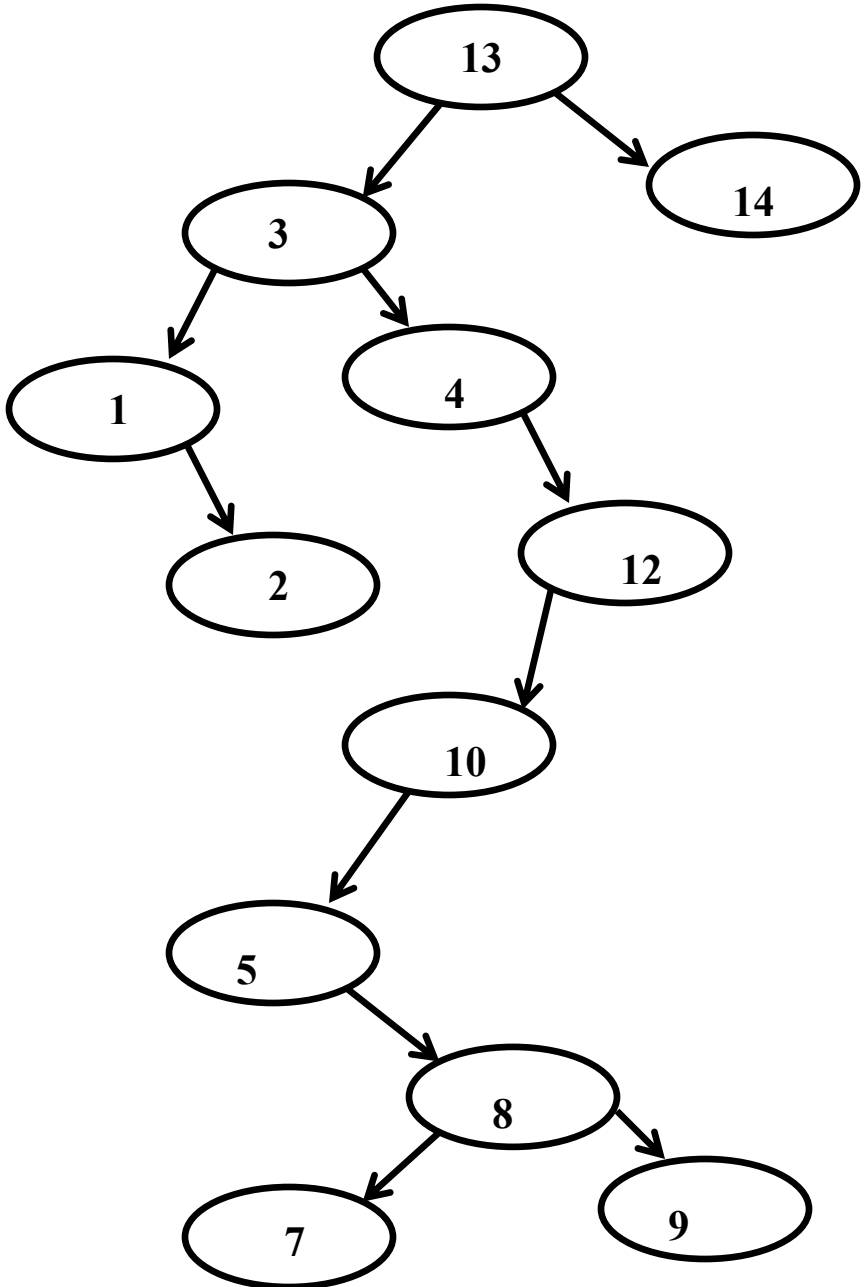

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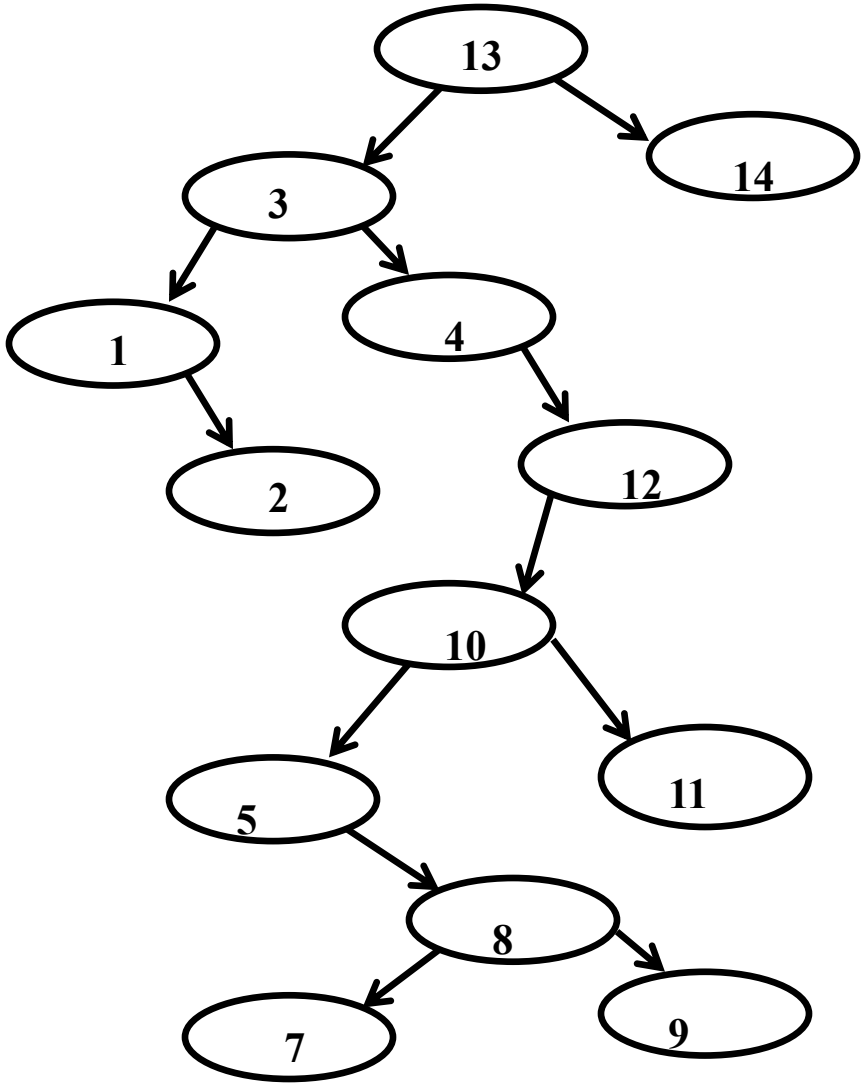

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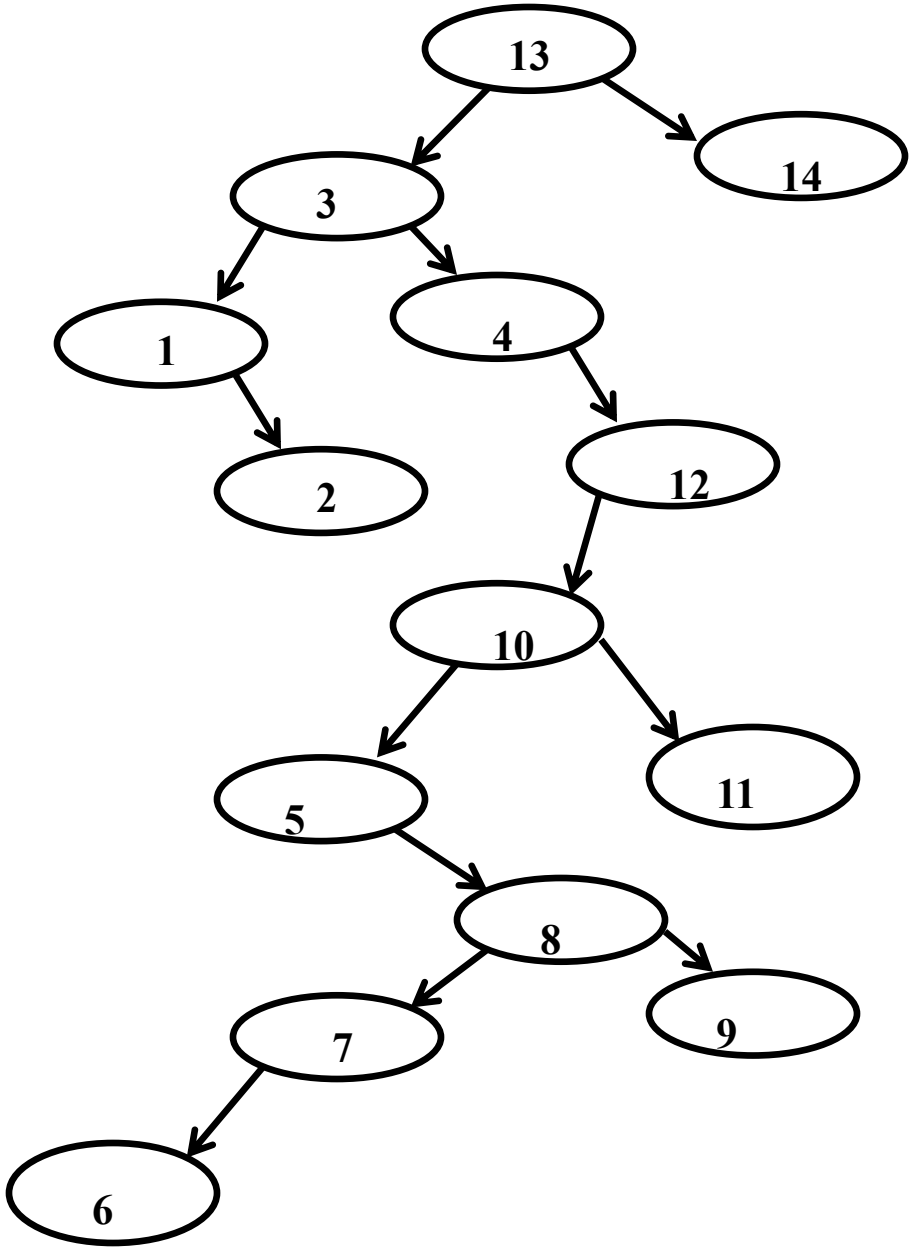

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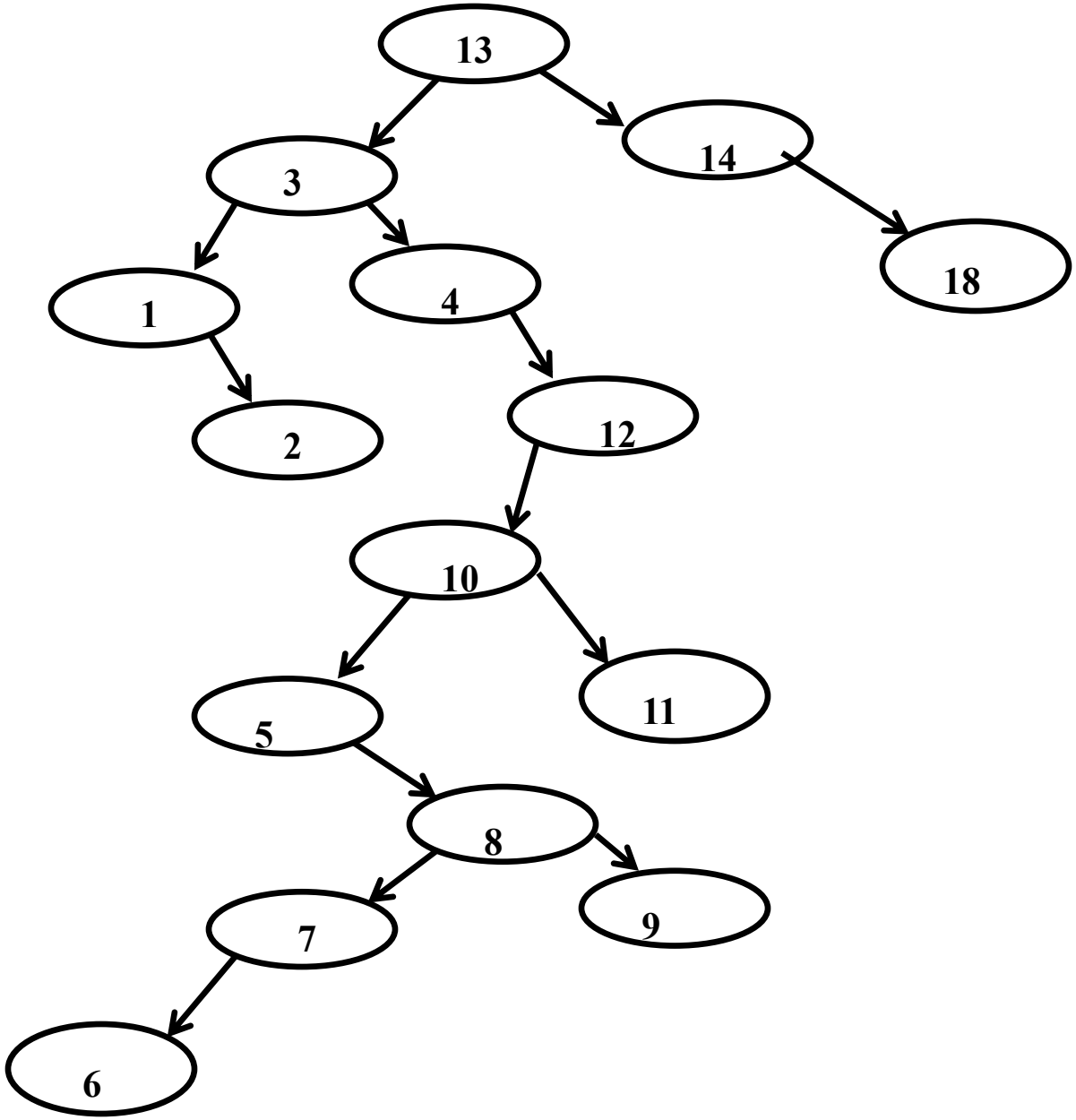
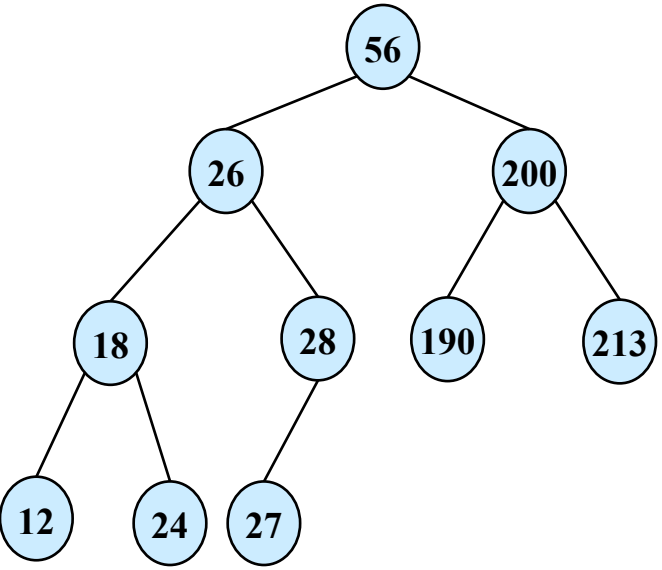


BST Insertion – Pseudocode




```
struct Tree *insert(struct Tree *root, int data)
{
If(root ==Null)
  {
    root=struct Tree* malloc(sizeof(struct Tree ));
    root->data=data;
    root->left=root->right=Null;
  }
else if(data<root->data)
    root->left=insert( root->left, data);
else if (data>root->data)
        root->right=insert( root->right, data);
}
Return root;
}
```

Tree-Delete (T, x)

if x has no children ♦ case 0

then remove x

if x has one child ♦ case 1

then make $p[x]$ point to child

if x has two children (subtrees) ♦ case 2

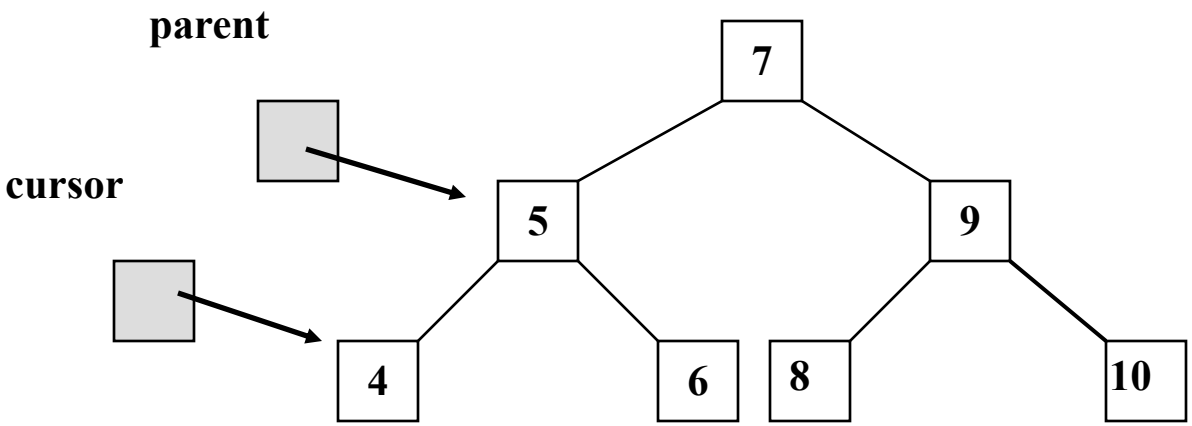
then swap x with its successor

perform case 0 or case 1 to delete it

$\Rightarrow$ TOTAL: $O(h)$ time to delete a node

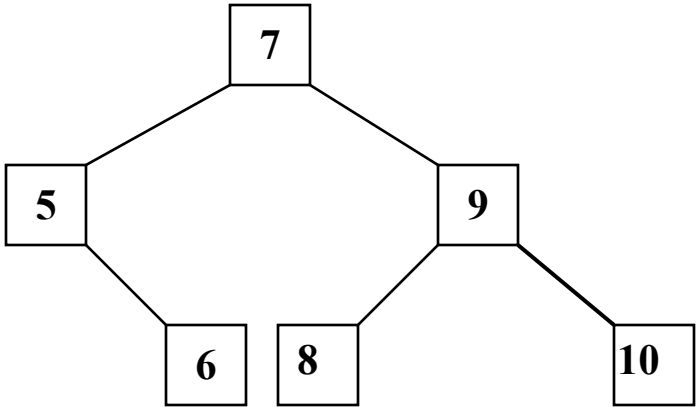
Removal in BST: Example

Case 0: **removing a node with 2 EMPTY SUBTREES**



Removing 4

replace the link in the parent with `null`

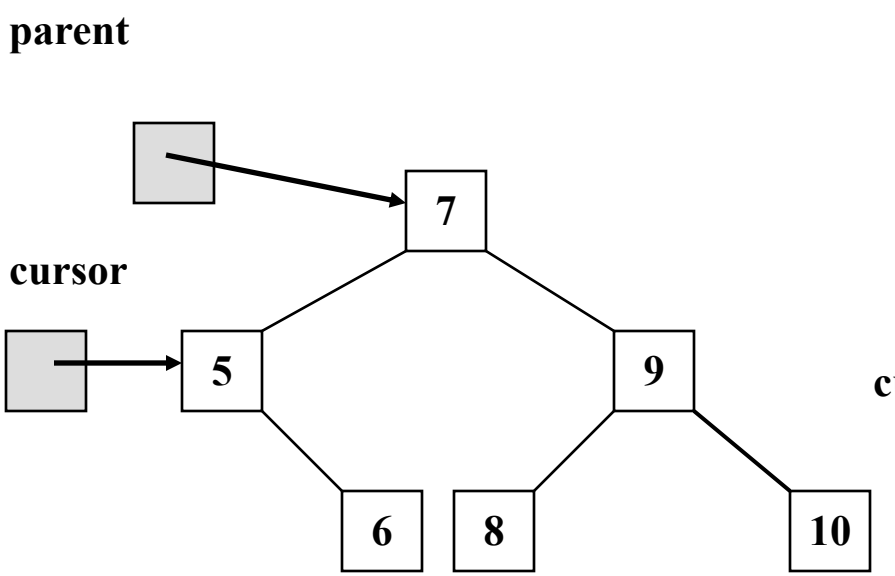


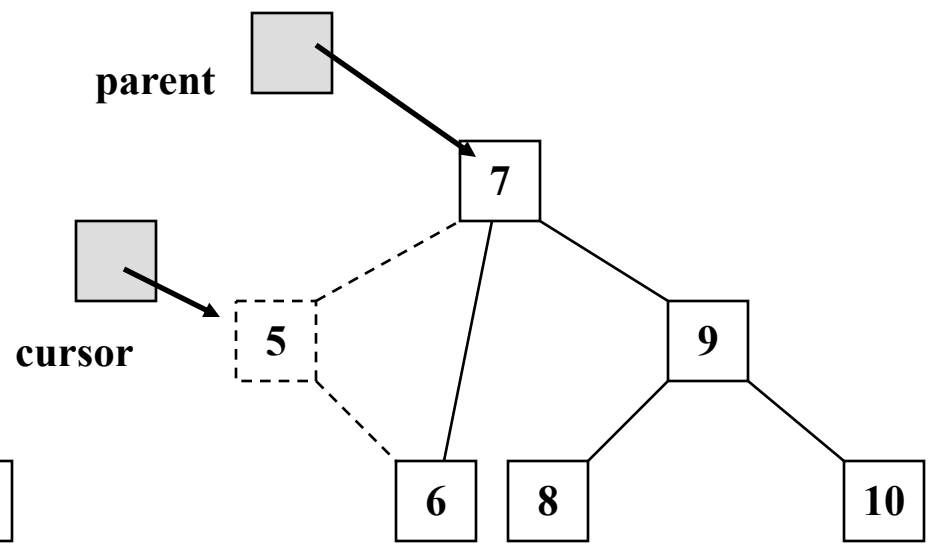
Removal in BST: Example

Case 1: **removing a node with 1 EMPTY SUBTREE**

the node has no left child:

link the parent of the node to the right (non-empty) subtree





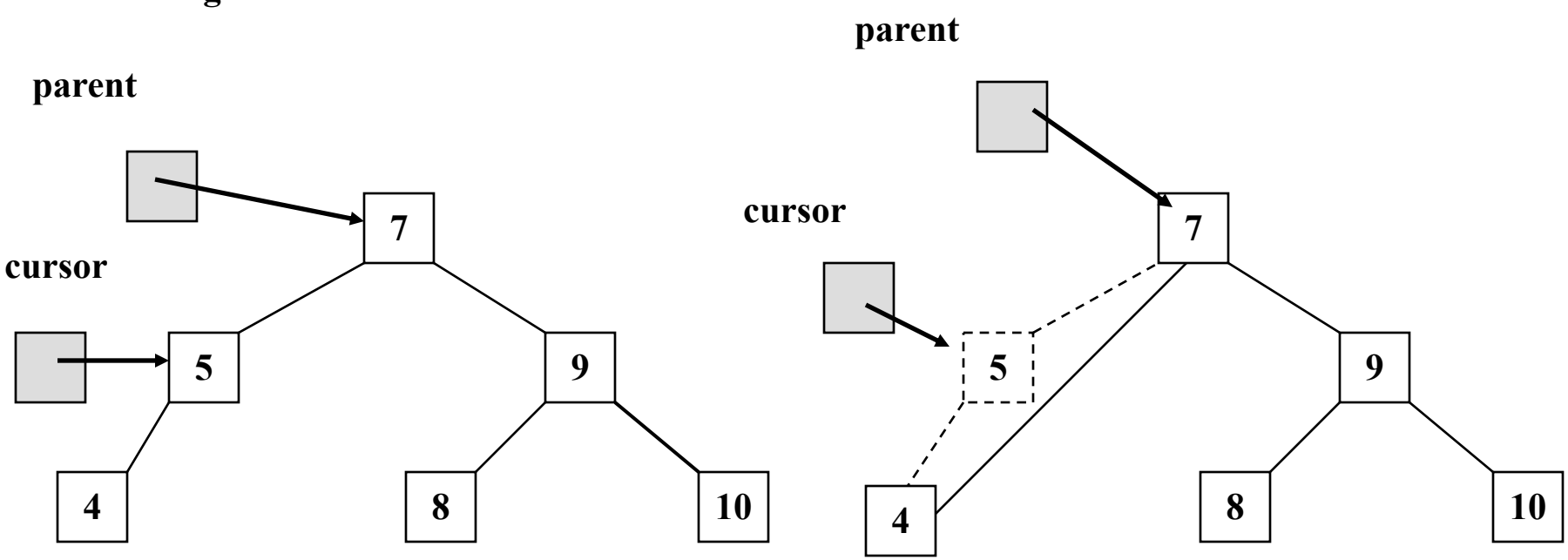
Removal in BST: Example

Case 1: removing a node with 1 EMPTY SUBTREE

the node has no right child:

link the parent of the node to the left (non-empty) subtree

Removing 5



Removal in BST: Example

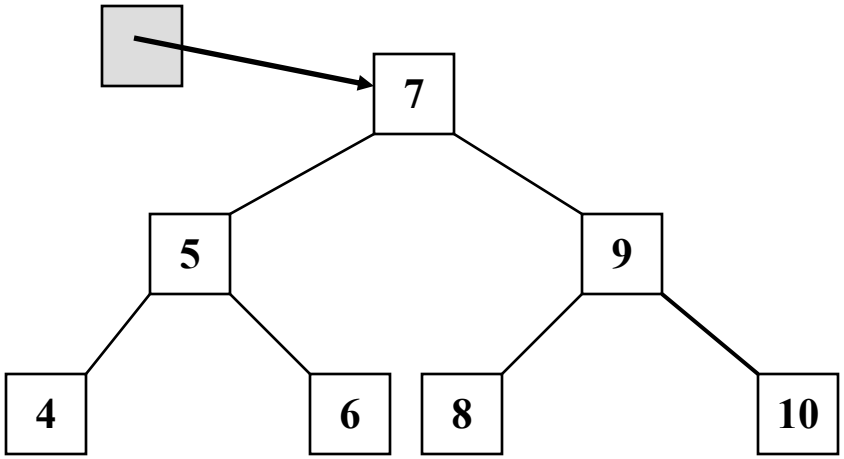
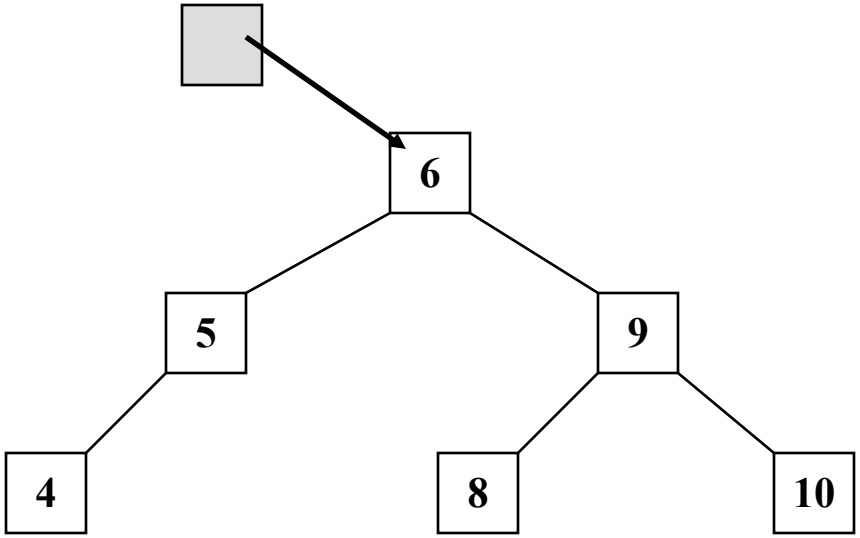
Case 2: **removing a node with 2 SUBTREES**

- **replace the node's value with the max value in the left subtree**
- **delete the max node in the left subtree**

What other element can be used as replacement?

Removing 7

cursor



cursor